

Do High Schools Shape College Choices?

Evidence from Centralized Assignment in Chile

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High schools may shape choices that test-score measures miss

We know that:

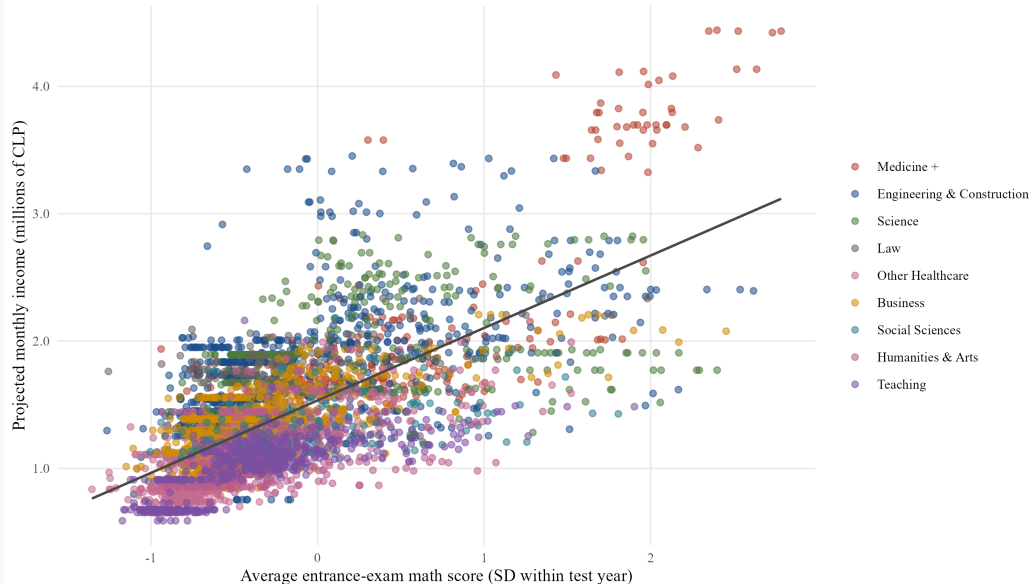
- Despite school quality being usually summarized by test scores, ample evidence of multiple dimensions. (Jackson 2018; Beuermann et al. 2023; Rose et al. 2022)
- College application and enrollment differ across groups even conditional on achievement, ample room for interventions. (Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025)
- Returns differ substantially across institutions and especially across fields. (Kirkeboen et al. 2016)

Research Question

Do high schools causally shape college choices, and are those effects captured by achievement value added?

Projected Program Income and Entrant Math Achievement

Each point is a program; 2022--2025 entering cohorts combined



Programs require at least 10 observed entrants. Health categories follow the Medicine + / Other Healthcare split used in the field-premium plot.

- I combine centralized high school assignment (with random tie-breaking) with information about higher educational choices.
- For each outcome I proceed in two steps:

1. Broad national panel

Estimate outcome-specific school value added using rich pre-treatment controls. Apply Empirical Bayes to reduce noise in estimates. (Chetty et al. 2014; Walters 2024) .

$$\hat{V}_j^{m,EB}$$

2. SAE lottery sample

Test whether an offer to a higher-value-added school (gains in $\hat{V}_j^{m,EB}$) causally improves the same outcome.

$$\varphi^m$$

Preview of Results

Outcome	Outcome mean	$P_{90} - P_{10}$ VA difference	φ	Implied causal gain $(P_{90} - P_{10}) \times \varphi$
Math	0.043	0.492 SD	0.884	0.435 SD
High-premium field	0.347	12.3 pp	0.803	9.9 pp

The final column is the lottery-validated causal gain implied by moving from a P_{10} to a P_{90} value-added school.

Five outcomes capture achievement and the college option entered

Achievement in higher-education exams

- standardized math score;
- standardized verbal score.

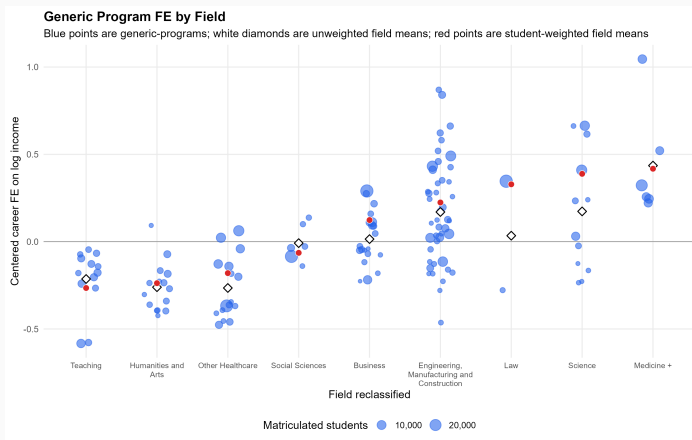
Higher-education choices

- **log projected income**, based on MiFuturo (avg. wage per program-institution reported by govt.)
- enrollment in a **high-premium field** (binary);
- enrollment in a **high-premium institution** (binary);

Outcome convention

Non-enrollment is coded as zero for the premium indicators and enters projected income at a minimum-wage proxy.

Projected-income premia vary sharply across fields



The high-premium-field outcome includes Science, Law, Engineering/Manufacturing/Construction, and Medicine+.

Centralized high-school assignment in Chile

- The *Sistema de Admisión Escolar* (SAE) began regionally in 2017 and became national in 2018. It covers public and private-subsidized schools, which enroll about 91% of students.
- Families applying to grade 9 submit ranked school lists. Student-proposing Deferred Acceptance combines rankings, vacancies, and priorities, using random tie-breakers within oversubscribed priority groups.
- I replicate the assignment algorithm and recover each applicant's school-offer probabilities through repeated counterfactual lottery draws.

Replication check

Using the original lottery numbers, the replicated mechanism reproduces more than 99% of observed first-round assignments in every 2018–2021 cohort.

Sample description

Value-added estimation

Grade-8 universe, 2017–2020 **944,510**



Complete value-added controls
Choice + income VA base **757,999**



Math or verbal score observed
Math + verbal VA **571,110**

Lottery IV

Timely grade-9 SAE applicants,
2018–2021 **409,737**



Applicants with assignment risk **226,195**



Complete grade-4 SIMCE
controls **139,317**



Exam takers with math or
verbal score
All five IV outcomes **95,705**

Step 1: Value Added Estimation

$$Y_i^m = \alpha^m + \lambda_{s(i)}^m + X_i' \Gamma^m + \mu_i^m, \quad \hat{V}_j^m = \hat{\lambda}_j^m - \bar{\lambda}_w^m.$$

- Estimate a separate school fixed effect per each outcome with universe.
- X_i includes grade-4 (national) achievement polynomials, family background, middle-school fixed-effect plus grades and attendance, cohort, gender, age, and home location.
- These measures remain observational: sorting not absorbed by X_i can contaminate $\hat{V}_j^m$.
- $\hat{V}_j^m$ are demeaned so average value-added estimate is zero.

Step 1.2: Empirical Bayes shrinks less precise school estimates more strongly

$$\hat{\tau}_m^2 = \max \left\{ \text{Var}_\omega(\hat{V}_j^m) - \sum_j \omega_j (se_j^m)^2, 0 \right\}, \quad \lambda_j^m = \frac{\hat{\tau}_m^2}{\hat{\tau}_m^2 + (se_j^m)^2}, \quad \hat{V}_j^{m,EB} = \lambda_j^m \hat{V}_j^m.$$

- Large, precisely estimated schools remain close to their original fixed effects.
- Small or noisy schools move toward the student-weighted mean of zero. (Walters 2024)
- Shrinkage improves the signal-to-noise ratio; it does not create a separate causal estimator.

Lottery identification: selection and randomization

Ranked lists capture selection

- Challenge of studying causal effects of schools is selection.
- Families reveal their 'types' through their ranked school lists.
- Rankings (and priorities) imply each applicant's assignment probabilities p_i .
- $E_i^{m,EB} = \sum_j p_{ij} \hat{V}_j^{m,EB}$ summarizes the school value added expected from the choices of that student 'type'.

Tie-breakers generate exogenous variation

- SAE sees only ranked lists and generates random tie-breakers.
- It does not see prior achievement, family background, potential outcomes, etc.
- Conditional on p_i , the tie-breaker makes the school offered quasi-random.

First Stage: Lottery offers shift attended-school value added

$$A_i^{m,EB} = a^m + \kappa^m O_i^{m,EB} + \rho_A^m E_i^{m,EB} + W_i' \delta_A^m + \nu_i^m.$$

Instrument and treatment

- $O_i^{m,EB} = \hat{V}_{Z_i}^{m,EB}$: value added of the first-round offered school.
- $A_i^{m,EB} = \hat{V}_{D_i}^{m,EB}$: value added of the school ultimately attended.

Conditioning on assignment risk

- $E_i^{m,EB} = \sum_j p_{ij} \hat{V}_j^{m,EB}$ summarizes the applicant's simulated offer probabilities.
- Conditional on p_i , random tie-breakers generate quasi-experimental variation in $O_i^{m,EB}$.
- W_i : cohort indicators, gender, age, and grade-4 SIMCE math and language scores.

Second Stage: Estimate pass-through using the lottery

$$Y_i^m = \alpha^m + \varphi^m A_i^{m,EB} + \rho^m E_i^{m,EB} + W_i' \delta^m + \eta_i^m, \quad A_i^{m,EB} \text{ instrumented by } O_i^{m,EB}.$$

- Y_i^m is the student outcome used to construct value added for outcome m .
- $E_i^{m,EB}$ absorbs expected value added from the student's assignment portfolio.
- W_i : cohort indicators, gender, age, and grade-4 SIMCE math and language scores.
- φ^m measures how much of a lottery-induced increase in attended-school value added passes through into a causal gain in outcome m .

Interpretation

$\varphi^m = 1$: one-for-one pass-through.

$\varphi^m = 0$: no causal content at the lottery margin.

Results: Value added predicts causal gains in all five outcomes

	Achievement		Higher-education choices		
	Math	Verbal	High-premium inst.	High-premium field	Log proj. income
φ^{EB}	0.884*** (0.073)	0.885*** (0.087)	0.497*** (0.111)	0.803*** (0.100)	0.311*** (0.103)
N	93,438	95,107	95,651	90,338	95,651

- Achievement pass-through is close to one despite the absence of immediately lagged test scores.
- School value added also predicts movement into higher-premium institutions, fields, and programs with higher projected income.

► Heterogeneity by baseline achievement

► Heterogeneity by gender

Robust standard errors in parentheses. All models control for expected EB value added and baseline covariates. *** $p < 0.01$.

Results: Estimates imply economically meaningful 90–10 school differences

Outcome	Outcome mean	$P_{90} - P_{10}$ VA difference	φ	Implied causal gain $(P_{90} - P_{10}) \times \varphi$
Math	0.043	0.492 SD	0.884	0.435 SD
Verbal	0.049	0.374 SD	0.885	0.331 SD
High-premium institution	0.142	23.4 pp	0.497	11.6 pp
High-premium field	0.347	12.3 pp	0.803	9.9 pp
Log projected income	13.893	0.183 log points	0.311	0.057 log points

The final column is the lottery-validated causal gain implied by moving from a P_{10} to a P_{90} value-added school.

Cross-outcome results: Math and field VA capture different margins

School value-added index	Math score		High-premium field	
	Estimate	SE	Estimate	SE
Math VA	0.884***	(0.073)	-0.184***	(0.054)
High-premium-field VA	0.099	(0.126)	0.803***	(0.100)

Each cell is a separate lottery IV. The attended-school VA in that row is instrumented by the corresponding offered-school VA; models control for its expected VA and baseline covariates.

Math VA → field choice

$P_{90} - P_{10}$ effect: **-9.1 pp**

Higher math VA does not proxy for movement
into high-premium fields.

Field VA → math score

$P_{90} - P_{10}$ effect: **+0.012 SD**

The estimate is small and statistically
insignificant ($p = 0.429$).

► Cross-outcome IV specification

Schools affect higher educational paths through multiple channels

- Observational value added predicts lottery-induced causal gains in achievement and the characteristics of higher educational choices.
- In the same way we can identify schools with large causal effects in mathematics achievement, we can identify schools that nudge students into high-premium fields and institutions.
- The effects are large enough to be relevant.
- Cross-outcome tests suggest that the effects on choices are not predicted by achievement: the effects operate through different margins.

Thanks!

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School quality references

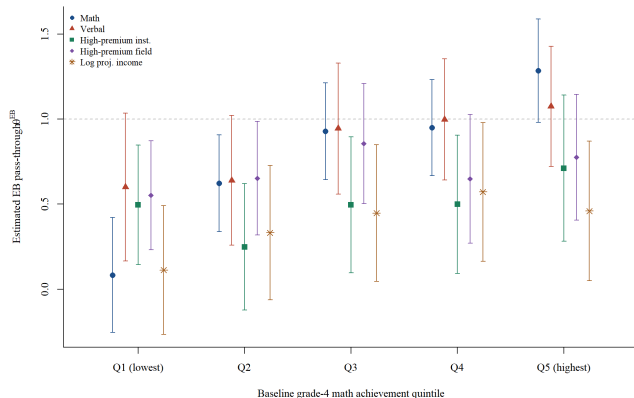
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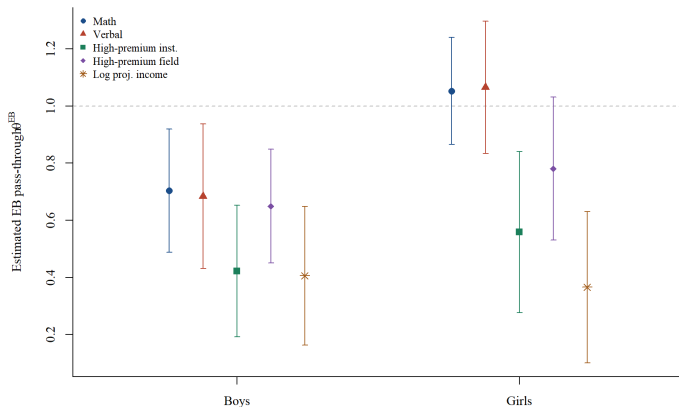
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Math pass-through is weakest for students with low baseline achievement



- Math: $\hat{\varphi}_{Q1} = 0.083$ versus $\hat{\varphi}_{Q3} = 0.929$.
- Difference: -0.846 ($p < 0.01$).
- Higher-education pass-through is stable across baseline math quintiles.

Achievement value added is more predictive for girls



- Math: girls' pass-through is 0.349 higher ($p = 0.017$).
- Verbal: girls' pass-through is 0.382 higher ($p = 0.029$).
- Gender differences are small and insignificant for the choice outcomes.